

Strategic Foundation for the Tail Expectation in Limit Order Book Markets

Shmuel Baruch and Lawrence R. Glosten¹

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¹Baruch: David Eccles School of Business, University of Utah, Salt Lake City, UT 84112; shmuel.baruch@business.utah.edu. Glosten: Columbia Business School, Columbia University, New York, NY 10027; lrg2@columbia.edu. This is a substantially revised version of an early paper, circulated under the name “Fleeting Orders.”

Abstract

In an environment with noise traders and informed traders trading on news, we model competition in schedules by liquidity suppliers quoting on a limit order book. We show that there is an equilibrium featuring quoters using mixed strategies; each quoter employs a step function, offering the same quantity of shares at random prices (and, of course, random bid prices). These random prices with their associated quantities form the market quotes and the depth of book, or price schedule. There are equilibria with a smaller number of quoters quoting a larger number of shares and equilibria with a larger number of quoters quoting a smaller number of shares. Considering a sequence of equilibria with the number of quoters becoming large, we establish that the stochastic equilibrium price schedule converges to the zero-profit deterministic competitive price schedule.

Keywords: competition in schedules, financial markets, limit orders, tail condition, flickering quotes

1 Introduction

Limit order book markets have all but replaced traditional exchanges. Accordingly, this market structure is of great interest to traders, regulators, and academics. A workhorse for studying limit order book markets is the conditional tail expectation condition (hereafter, the tail condition). In a market in which quoters face informed trade, this is a zero-profit condition which states that limit prices equal the conditional expected value of the asset, given their execution.¹ In certain economic environments, [Back and Baruch \(2013\)](#) and [Biais, Martimort, and Rochet \(2013\)](#) show that the tail condition is, in fact, the limit equilibrium as the number of strategic liquidity suppliers becomes large. The goal of this paper is to extend these authors' results. Specifically, we study the time-honored environment in which the liquidity demander is either a noise trader with *inelastic demand* or an informed trader.²

Under a sufficient degree of adverse selection, [Back and Baruch \(2013\)](#) and [Biais, Martimort, and Rochet \(2013\)](#) find equilibrium in pure strategies in which liquidity suppliers employ continuous schedules. The inelastic demand of noise traders, analyzed in this paper, implies that the degree of adverse selection is low and that, as a result, there is no equilibrium in pure strategies. To see this, consider a liquidation value that has unbounded support, and assume that the noise demand has least upper bound Z . In a pure strategy equilibrium, no more than Z shares can be offered. A profitable deviation for anyone who offers shares is to offer those shares at infinity. This deviation generates unbounded expected profits. Clearly, this argument is an artifact of the assumption that noise traders are willing to pay any price (i.e., there is a low degree of adverse selection at high prices). However, this is the assumption that the literature favors because a no-trade equilibrium is avoided: When

¹There is a literature that studies the limit order book market while abstracting from information asymmetry. In this literature, the tail condition has no role. Important papers include [Seppe \(1997\)](#), [Parlour \(1998\)](#), [Foucault \(1999\)](#), [Parlour and Seppe \(2003\)](#), [Foucault, Kadan, and Kandel \(2005\)](#), and [Roşu \(2009\)](#).

²This is by far the most common setup in the literature. Among the many papers that use this environment are [Glosten and Milgrom \(1985\)](#), [Diamond and Verrecchia \(1987\)](#), [Easley and O'hara \(1987\)](#), and [Colliard \(2016\)](#).

noise traders’ demands are price inelastic, liquidity suppliers can always recoup their losses to informed traders. In this paper, we bridge the gap left by [Back and Baruch \(2013\)](#) and [Biais, Martimort, and Rochet \(2013\)](#) by focusing on mixed strategy equilibria. Although we do not rule out the continuous schedule abstraction used in those papers, the equilibrium schedules in this paper are random step functions that can be implemented using plain limit orders.

[Kyle \(1989\)](#), [Vives \(2011\)](#), and [Rostek and Weretka \(2012\)](#) also study competition in schedules with private information. However, these papers focus on a trading platform that uses a single price to clear the market. Although many financial markets do use a single price auction to open the trading day, we focus on a limit order market. [Dennert \(1993\)](#) models a similar environment to ours and finds the equilibrium in mixed strategies. In that paper, however, liquidity suppliers compete in prices (quantities are exogenous).

Our model is an example of a game with discontinuous payoffs. Here, the discontinuity in payoffs is due to the discriminatory nature of limit order markets (“pay your bid” auction): Offering shares at a price marginally lower than the opponent’s offering price is dramatically different than offering the shares at a price marginally higher, because the former implies a greater chance of trading with uninformed noise traders.

Interestingly, the equilibria we study here share similarities with those found in the Bertrand–Edgeworth literature. Specifically, in that literature, producers choose capacities in pure strategies and randomize the prices (see [Tirole \(1988\)](#)). While in our model competition is in schedules, in the equilibria we find, liquidity suppliers choose deterministic quantities and randomize prices. Our equilibria are similar to those in the Bertrand–Edgeworth literature in that the entire demand may not be satisfied at low prices.

An important goal of the Bertrand–Edgeworth literature is to provide a strategic foundation for competitive pricing. Specifically, [Allen and Hellwig \(1986a\)](#), [Allen and Hellwig \(1986b\)](#),

and [Vives \(1986\)](#) demonstrate the convergence of mixed strategy equilibria, as the number of suppliers becomes large, to the competitive deterministic price.³ Our paper extends this literature to a case in which competition is in schedules; the demand is random; suppliers face adverse selection; and the limit is a deterministic competitive schedule, rather than a single competitive price.

In the reminder of this section, we discuss the tail condition and its importance in the limit order book literature. Orders are anonymous, so the only attribute a market order carries is its size. Price priority means that the execution of a limit order placed away from the market occurs only when the size of the incoming marketable order is sufficiently large to walk through the book until it hits the limit order. Therefore, in the tail condition, the conditioning is that the size of the incoming marketable order must be larger than the number of shares quoted at the limit order or better (lower in the case of an offer, higher in the case of a bid).

To fix the intuition, let us suppose that a market has offers for 500 shares at 19.51, 400 shares at 19.52, and 300 shares at 19.53. The rules of the exchange require price priority, so that all the shares at 19.52 will transact if and only if a marketable buy order sent to the exchange is for 900 shares or more. Thus, this offer of 19.52 should satisfy the condition that the expected value of the security, conditional on a marketable buy order of 900 shares or more, is no greater than 19.52. Similarly, an additional share will be added to the 300 at 19.53 only if the expected value of the security conditional on a marketable buy order of more than 1200 shares is no greater than 19.53. The zero-profit tail condition results from competition driving the “no greater than” phrases above to “equal to.”

Two aspects of the operation of the limit order market lead to this condition. First, as noted

³Interestingly, [Dennert \(1993\)](#) demonstrates the existence of a sequence of equilibria that do not converge to the competitive price. However, by assumption, the noise demand is fixed at one unit, whereas the liquidity supply in a market with n liquidity suppliers is fixed at n units.

above, the market observes price priority. Second, limit orders are firm in that they cannot be canceled in the midst of a marketable order execution.

The tail condition is useful for understanding important features of stock trading. [Rock \(1990\)](#) uses the tail condition to demonstrate that specialist price improvement may have worsened market quality. Rock observed that, when we had specialists on the NYSE, a specialist could see the size of the incoming market order and, to the extent that large orders are more likely to be informed, could gain from providing price improvement to small orders even if the quotes he had to improve upon satisfied the zero-profit condition. According to empirical work by [Ready \(1999\)](#), specialists did indeed tend to choose the most profitable orders for price improvement, thereby imposing significant adverse selection costs on the limit order book.⁴

In principle, this imposition of adverse selection costs could lead to a breakdown in the limit order book. Suppose the specialist offered price improvement on orders of up to 200 shares, but the limit order providers believed that all orders hit the book. The limit order users would then observe that they received no small, safe orders and would revise the quotes so that the market bid and offer would feel comfortable handling more than 200 shares. This widening of the quote would then lead the specialist to handle, for example, 400 shares. This, in turn, would lead to a widening of the spread, and so on.

A similar analysis applies to the practice of payment for order flow. Over the counter (OTC) dealers are profitably able to purchase the right to execute orders placed with the brokers who are most likely to handle retail order flow. According to Rule 611 of Regulation National Market System (Reg NMS), these dealers cannot execute the orders at prices inferior to the National Best Bid and Offer (NBBO), and in fact, some offer price improvement (allbeit at one one-hundredth of a penny). They can do so if the market bid and offer satisfy the

⁴[Seppi \(1997\)](#) notes that even without informative order flow, a specialist imposes an execution risk on limit orders placed in the book.

tail condition. The OTC dealers are confident that they are transacting small retail orders, whereas the market bid and offer must condition on servicing an order of any size, from small and safe to large and dangerous. As in the analysis by [Rock \(1990\)](#) of price improvement by specialists, payment for order flow can have a negative impact on market liquidity. With small, safe orders handled OTC, the mix of orders going to the exchanges becomes more dangerous.⁵

[Glosten \(1994\)](#) uses the tail condition to demonstrate the existence of a positive small-trade spread and shows that a consolidated market resembles a limit order book market. Specifically, because liquidity demanders can split their orders among different exchanges, competition among exchanges mimics the competition among liquidity suppliers in a limit order book. It is this property that makes the condition useful for examining competition among multiple exchanges, even though it formally applies to a centralized limit order book (CLOB).

[Back and Baruch \(2007\)](#) show that the tail condition can also be the pricing rule in trading floors. In the worked-order equilibrium they find, large traders split their orders into small pieces. Thus, even though floor traders can condition their quotes on the size of an incoming order, they cannot be sure that the order is not part of a large worked order.

The rest of this paper is organized as follows. In Section 2 we describe the model and define the tail condition. In Section 3 we demonstrate the existence of a sequence of Nash equilibria. In Section 4 we prove a convergence result. In Section 5 we provide some examples. In Section 6 we show how the model can be extended. In Section 7 we conclude.

⁵For a legal analysis of this topic, see [Haeberle \(2015\)](#). For a different rationale for payment for order flow in the context of a dealer market, see the dynamic model in [Parlour and Rajan \(2003\)](#).

2 The Limit Order Book and the Tail Condition

We consider a market for a single risky asset. The market is organized as a pure limit order book with the usual price priority: An incoming marketable order walks the book, picking off outstanding limit orders at their limit prices.⁶

The timing is as follows. First, uninformed risk-neutral liquidity suppliers simultaneously submit limit orders. Next, a single liquidity demander submits a marketable order. Finally, the asset liquidates and pays the liquidation value $\tilde{v}$. The liquidation value is drawn from either a discrete or a continuous distribution, bounded or unbounded. However, we require that $\tilde{v}$ has a finite expected value, and we denote by V the least upper bound of $\tilde{v}$.⁷

We assume that any real number is a feasible price. We focus on equilibria in which buy and sell limit orders of liquidity suppliers do not cross—in other words, uninformed liquidity suppliers do not trade among themselves. In such equilibria, we can study the offer side separately from the bid side; hereafter, for simplicity of exposition, we focus on the offer side of the book.

The liquidity demander can be one of two types, a news trader (with probability μ) or a noise trader (with probability $1 - \mu$). A news trader knows $\tilde{v}$ and picks off all offers at prices smaller than $\tilde{v}$.⁸ A noise trader submits a market order of size $\tilde{z}$. If $\tilde{z} > 0$, then the order is a buy order. We assume $\tilde{z}$ is a finite integer (i.e., whole lot) random variable. Let $\mathcal{Z}^+$ denote the strictly positive part of the support, and let Z be the maximal element in $\mathcal{Z}^+$.

⁶A marketable order is an order that can be executed upon submission. Any buy (respectively, sell) limit order with a price higher (lower) than the ask price (bid price) is marketable. In particular, market orders are marketable.

⁷If $\tilde{v}$ is unbounded, then V is infinity.

⁸A limit buy order with limit price $\tilde{v}$ and a sufficiently large quantity can pick off all offers. The news trader also picks off all bids at prices higher than $\tilde{v}$, but because we focus on the offer side, this does not concern us.

We can trivially write

$$\mathcal{Z}^+ \subset \{i : i \in \mathbb{N}\} \quad (1)$$

Next, we turn our attention to the liquidity suppliers. We represent a collection of offers using a nondecreasing marginal price schedule $P(\cdot)$, with the interpretation that the q th unit, $q > 0$, is offered at $P(q)$. Thus, if $P(\cdot)$ stands for the entire collection of outstanding offers in the book, and $\tilde{q}$ stands for the size of the incoming order, then

$$\begin{aligned} \forall q > 0, \quad \{q \leq \tilde{q}\} &= \{q \leq \tilde{q}, \text{demander is news}\} \cup \{q \leq \tilde{q}, \text{demander is noise}\} \\ &= (\{\text{demander is news}\} \cap \{P(q) < \tilde{v}\}) \cup (\{\text{demander is noise}\} \cap \{q \leq \tilde{z}\}) \end{aligned}$$

The four events are disjoint, and we can write

$$\text{Prob}(q \leq \tilde{q}) = \mu \text{Prob}(P(q) < \tilde{v}) + (1 - \mu) \text{Prob}(q \leq \tilde{z})$$

The aggregate expected profits of the liquidity suppliers are then

$$\begin{aligned} E \int_0^\infty I_{\{q \leq \tilde{q}\}} (P(q) - \tilde{v}) dq \\ &= \int_0^\infty \mu E I_{\{P(q) < \tilde{v}\}} (P(q) - \tilde{v}) + (1 - \mu) E I_{\{q \leq \tilde{z}\}} (P(q) - \tilde{v}) dq \\ &= \int_0^\infty \alpha(P(q)) + \beta(P(q)) \text{Prob}(q \leq \tilde{z}) dt \end{aligned}$$

where

$$\begin{aligned} \alpha(p) &:= \mu E I_{\{p < \tilde{v}\}} (p - \tilde{v}) \\ \beta(p) &:= (1 - \mu)(p - E\tilde{v}) \end{aligned} \quad (2)$$

The term $\alpha(P(q))$ represents the expected gains from trading with an informed liquidity demander (these gains are negative). The term $\beta(P(q)) \text{Prob}(\tilde{z} \geq q)$ represents the expected gains from trading with an uninformed liquidity demander.

A marginal price schedule is called *competitive*, and we denote it by $P^c(\cdot)$, if the integrand is identically zero:

$$\forall q \in (0, Z], \quad \alpha(P^c(q)) + \beta(P^c(q)) \text{Prob}(q \leq \tilde{z}) = 0 \quad (3)$$

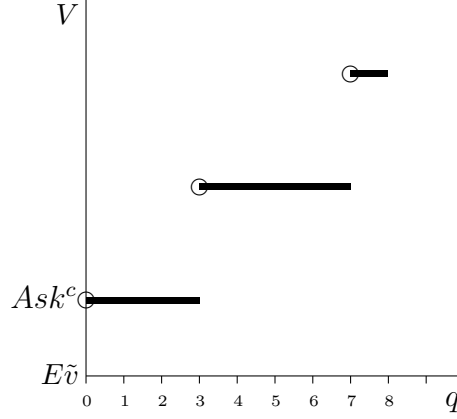


Figure 1: Graph of a competitive marginal price schedule, $P^c(\cdot)$: In this example, $Z = 8$, and $\mathcal{Z}^+ = \{3, 7, 8\}$. There are three offering prices, each of which corresponds to an element of $\mathcal{Z}^+$. All offering prices are strictly higher than $E\tilde{v}$ and strictly lower than V .

Thanks to the continuity and monotonicity of $\alpha(\cdot)$ and $\beta(\cdot)$, the existence and uniqueness of the competitive price schedule are guaranteed. Moreover, (i) $P^c(\cdot)$ is a left continuous step function, and (ii) the set of discontinuity points is $\mathcal{Z}^+ \setminus \{Z\}$. These properties of $P^c(\cdot)$ are inherited from the term $\text{Prob}(q \leq \tilde{z})$. Figure 1 shows an example of a competitive marginal price schedule.

We let Ask^c be the price that satisfies

$$\alpha(\text{Ask}^c) + \beta(\text{Ask}^c) \text{Prob}(0 < \tilde{z}) = 0 \quad (4)$$

Then, $\text{Ask}^c = \lim_{q \searrow 0} P^c(q)$ is the lowest price at which units are offered.

The competitive marginal price schedule is typically expressed in terms of the upper tail condition:

$$\forall q \in (0, Z], \quad \left(P^c(q) - E[\tilde{v} | q \leq \tilde{q}] \right) \text{Prob}(q \leq \tilde{q}) = 0$$

which is simply a rearrangement of (3). Hereafter, we refer to (3) as the *tail condition*.

3 Sequence of Nash Equilibria

We assume that the number of potential liquidity suppliers is unbounded and that each of them is strategic and can post an arbitrary collection of offers (i.e., a nondecreasing marginal price schedule). There are many equilibria, but we are interested in a sequence of zero-rent, mixed strategy equilibria, which we parameterize by m . Neither of the equilibria in the sequence satisfies the tail condition, but in the next section we show that their limit does.⁹

The zero-rent condition means that a supplier's expected losses to the informed liquidity demander equal the supplier's expected gains from the uninformed demander. The failure to satisfy the tail condition means

$$\begin{array}{ccc} \sum_{\text{suppliers}} \text{expected losses to informed} & \neq & \text{losses to informed under tail condition} \\ \parallel & & \parallel \\ \sum_{\text{suppliers}} \text{expected gains from uninformed} & \neq & \text{gains from uninformed under tail condition} \end{array}$$

In other words, each element in the summation over losses equals the corresponding element in the summation over gains. The sum, however, does not equal the competitive outcome.

Theorem 3.1. *Let $m \in \mathbb{N}$, and let*

$$\begin{aligned} n_m &\equiv mZ \\ \Delta_m &\equiv 1/m \end{aligned} \tag{5}$$

1. *There is a strictly increasing continuous distribution function $M_m(\cdot)$ with support*

⁹It is immediately apparent that the tail condition cannot be a Nash equilibrium, because any supplier who offers shares can benefit by offering the same shares at a slightly higher price.

$[\text{Ask}^c, V]$ that satisfies

$$\forall p \in [\text{Ask}^c, V], \quad \alpha(p) + \beta(p) \text{Prob}(\tilde{M}_m(p) < \tilde{z}/Z) = 0 \quad (6)$$

where $\tilde{M}_m(p)$ is the proportion of successes in some arbitrary random sample of n_m Bernoulli trials.¹⁰

$$I_{\{\tilde{p}_1 \leq p\}}, I_{\{\tilde{p}_2 \leq p\}}, \dots, I_{\{\tilde{p}_{n_m} \leq p\}}, \quad \tilde{p}_i \text{ i.i.d. with distribution function } M_m(\cdot)$$

2. Let $1 + n_m$ liquidity suppliers offer a block of Δ_m at i.i.d. prices with the common distribution function $M_m(\cdot)$. All other potential liquidity suppliers are nonactive. Then

(a) An active liquidity supplier has no profitable deviation.

(b) A nonactive liquidity supplier has no profitable deviation.

The proof of the theorem is in the Appendix. The presence of the nonactive liquidity suppliers is not necessary to support the equilibrium. The active traders' strategies also form an equilibrium in an economy consisting of $1 + n_m$ potential liquidity suppliers, all of whom are active.

We refer to (6) as the *Nash condition*. Note the key difference between the tail condition and the Nash condition: The former requires that shares be offered at prices that yield zero expected profits, and the latter is a standard game theory condition on the mixing

¹⁰For a given p ,

$$\tilde{M}_m(p) = \frac{1}{n_m} \sum_{i=1}^{n_m} I_{\{\tilde{p}_i \leq p\}}, \quad \tilde{p}_i \text{ i.i.d. } M_m(\cdot)$$

$\tilde{M}_m(p)$ is known as the proportion of successes or the sample mean. Whatever name we attach to it, the notation is meant to remind the reader that we are in the realm of the law of large numbers:

$$E\tilde{M}_m(p) = M_m(p)$$

and

$$\text{Var}(\tilde{M}_m(p)) = \frac{M_m(p)(1 - M_m(p))}{n_m} \leq \frac{1}{4n_m}$$

distribution of a player's opponents. The condition requires that the opponents' mixing distribution be such that a supplier is indifferent as to where to offer the block of shares. In the remainder of the section, we discuss why the Nash condition is a necessary condition for a zero-rent equilibrium. In the proof of the theorem, we show that the condition is sufficient.

Note that the support of the mixed strategy distribution, $M_m(\cdot)$, must be no wider than the interval $[\text{Ask}^c, V]$. Indeed, selling the block Δ_m at prices lower than the competitive asking price generates a loss, and selling at prices higher than V generates gains—the latter because adverse selection is nonexistent at prices higher than V . Avoiding losses is self-reinforcing, but giving up on gains is not. The equilibrium in Theorem 3.1 satisfies the condition

$$n_m \Delta_m = Z$$

which implies that any arbitrary group of n_m active liquidity suppliers meets the entire noise demand at prices lower than V . Therefore, no single liquidity supplier can sell at prices strictly higher than V .

The Nash condition states that any arbitrary group of n_m active liquidity suppliers should not leave profits for the remaining active supplier to grab. To see this, let $\tilde{p}_1, \tilde{p}_2, \dots, \tilde{p}_{n_m}$ be the random prices at which the members of the group post their offers. These prices are i.i.d. with common distribution function $M_m(\cdot)$. The probability that the $(n_m + 1)$ th active supplier can trade with a noise trader at p , given that the demander is a noise trader, is the probability that the group does not entirely consume the noise order:¹¹

$$\begin{aligned} \text{Prob} \left(\sum_{i=1}^{n_m} \Delta_m I_{\{\tilde{p}_i < p\}} < \tilde{z} \right) &= \text{Prob} \left(\frac{1}{n_m} \sum_{i=1}^{n_m} n_m \Delta_m I_{\{\tilde{p}_i < p\}} < \tilde{z} \right) \\ &= \text{Prob} \left(\frac{1}{n_m} \sum_{i=1}^{n_m} Z I_{\{\tilde{p}_i < p\}} < \tilde{z} \right) = \text{Prob}(\tilde{M}_m(p) < \tilde{z}/Z) \end{aligned}$$

If $P(\cdot)$ is an arbitrary price schedule that the $(n_m + 1)$ th active supplier employs, then the

¹¹For the second equality, we use the identity $n_m \Delta_m = Z$. For the third equality, we use the definition of $\tilde{M}_m(p)$.

Nash condition requires that

$$\forall q \in (0, \Delta], \quad \alpha(P(q)) + \beta(P(q)) \text{Prob} \left(\sum_{i=1}^{n_m} \Delta_m I_{\{\tilde{p}_i < P(q)\}} < \tilde{z} \right) = 0$$

In other words, similar to the tail condition, the expected gains from trading with an informed liquidity demander (which are negative) and the expected gains from trading with an uninformed liquidity demander add up to zero.

4 Convergence

In this section, endowed with the sequence of equilibria, we prove a convergence result. In Theorem 4.3, below, we show that for all $q \notin \mathcal{Z}^+$, the sequence of random prices at which the q unit is offered at the m th equilibria converges in distribution/probability to the same price at which the q unit is offered under the tail condition.¹²

The roadmap of the proof is as follows. Write

$$u(p, q) \equiv \alpha(p) + \beta(p) \text{Prob}(q < \tilde{z})$$

Consider $q \notin \mathcal{Z}^+$, so $\text{Prob}(q \leq \tilde{z}) = \text{Prob}(q < \tilde{z})$, and the tail condition becomes $u(P^c(q), q) = 0$. We could use the law of large numbers (LLN) to conclude that, at the limit, the Nash condition becomes $u(p, ZM_\infty(p)) = 0$ for some real number $M_\infty(p)$. We might then hope that $ZM_\infty(p)$ is the inverse of $P^c(q)$, and if this is the case, our convergence result is all but complete. However, $P^c(\cdot)$ is a noninvertible step function.¹³

That said, the above is a good description of our convergence argument. In Lemma 4.1, we prove the variant of the LLN we need. In Lemma 4.2, we prove that $ZM_\infty(p)$ converges to an “inverse” of $P^c(\cdot)$; that is, if $P^c(\cdot)$ had an inverse, then this is exactly what Lemma 4.2

¹²Convergence in probability implies convergence in distribution. When the limit is a constant—here, $P^c(q)$ —the reverse is also true; in other words, convergence in distribution implies converges in probability.

¹³Put differently, $\partial u / \partial q \equiv 0$, and we cannot use the implicit function theorem.

would imply. Theorem 4.3 delivers the convergence of the random marginal prices in the Nash equilibrium to the deterministic marginal prices that satisfy the tail condition.

Lemma 4.1. *Fix p . If $M_{m_k}(p)$ is a convergent subsequence, then $\tilde{M}_{m_k}(p)$ converges in distribution to the same limit:*

$$\lim \text{Prob}(\tilde{M}_{m_k}(p) \leq x) = \begin{cases} 0 & \lim_{m_k \rightarrow \infty} M_{m_k}(p) > x \\ 1 & \lim_{m_k \rightarrow \infty} M_{m_k}(p) < x \end{cases}$$

The proof is in the Appendix. It need not concern us that the lemma is mute about what happens when $\lim_{m_k \rightarrow \infty} M_{m_k}(p) = x$. To prove the next lemma, we contrast the tail condition with the Nash condition.

Lemma 4.2. *Fix $q \in (0, Z)$ with $q \notin \mathcal{Z}^+$. We have*

$$\lim \text{Prob}(\tilde{M}_m(p) \leq q/Z) = \begin{cases} 0 & p > P^c(q) \\ 1 & p < P^c(q) \end{cases}$$

The proof is in the Appendix.

Let $\tilde{P}^m(\cdot)$ denote the random aggregate price schedule in the m th equilibrium. For each m , we have

$$\begin{aligned} & \{\tilde{P}^m(q) \leq p\} \\ & \quad \parallel \\ & \left\{ \sum_{i=1}^{n_m} \Delta_m I_{\{\tilde{p}_i \leq p\}} \geq q \right\} \subseteq \left\{ \sum_{i=1}^{1+n_m} \Delta_m I_{\{\tilde{p}_i \leq p\}} \geq q \right\} \subseteq \left\{ \sum_{i=1}^{n_m} \Delta_m I_{\{\tilde{p}_i \leq p\}} + \Delta_m \geq q \right\} \\ & \quad \parallel \qquad \qquad \qquad \parallel \\ & \left\{ \tilde{M}_m(p) \geq q/Z \right\} \qquad \qquad \qquad \left\{ \tilde{M}_m(p) \geq q/Z - \frac{1}{n_m} \right\} \end{aligned}$$

Thus,

$$\text{Prob}(\tilde{M}_m(p) \geq q/Z) \leq \text{Prob}(\tilde{P}^m(q) \leq p) \leq \text{Prob}\left(\tilde{M}_m(p) \geq q/Z - \frac{1}{n_m}\right) \quad (7)$$

Theorem 4.3. *Fix $q \in (0, Z)$ with $q \notin \mathcal{Z}^+$. Then $\tilde{P}^m(q)$ converges in distribution to $P^c(q)$:*

$$\lim_{m \rightarrow \infty} \text{Prob}(\tilde{P}^m(q) \leq p) = \begin{cases} 0 & p < P^c(q) \\ 1 & p > P^c(q) \end{cases}$$

The proof is in the [Appendix](#) and is based on a sandwich argument applied to (7). Theorem 4.3 has nothing to say about $q > Z$. That said, in the competitive equilibrium a total of Z units are offered, whereas in the Nash equilibrium the total is $Z + \Delta_m \rightarrow Z$.

5 Examples

In this section, we provide two examples in which it is possible to compute explicitly the mixing distribution.

Example 1. $\mathcal{Z}^+$ is a singleton: Because Z is the only element in $\mathcal{Z}^+$, $\text{Prob}(Z = \tilde{z}) = \text{Prob}(0 < \tilde{z})$, and

$$\begin{aligned} \text{Prob}(\tilde{M}_m(p) < \tilde{z}/Z) &= \text{Prob}(0 < \tilde{z}) \times \text{Prob}(\tilde{M}_m(p) < 1) \\ &= \text{Prob}(0 < \tilde{z}) \times \left(1 - \text{Prob}(\tilde{M}_m(p) = 1)\right) = \text{Prob}(0 < \tilde{z}) \times (1 - (M_m(p))^{n_m}) \end{aligned}$$

Inserting this result into the Nash condition, and recalling $n_m = mZ$, we obtain

$$\forall p \in (\text{Ask}^c, V), \quad M_m(p) = \left(1 + \frac{\alpha(p)}{\beta(p) \text{Prob}(0 < \tilde{z})}\right)^{1/mZ}$$

where Ask^c , as always, is given by (4).

Example 2. $m = 1$ and the conditional distribution of $\tilde{z}$, given $\{0 < \tilde{z}\}$, is $\mathcal{U}\{1, 2, \dots, Z\}$.

In this example, we consider only $m = 1$; thus, $n_1 = Z$. Our goal is to compute $M_1(\cdot)$ explicitly.

For each p , the support of $\tilde{M}_1(p)$ is $\{0/Z, 1/Z, \dots, Z/Z\}$, so¹⁴

$$\sum_{i=1}^Z \text{Prob}(\tilde{M}_1(p) \geq i/Z) = Z E \tilde{M}_1(p) = Z M_1(p)$$

Thus,

$$\sum_{i=1}^Z \text{Prob}(\tilde{M}_1(p) < i/Z) = Z(1 - M_1(p))$$

Now, $\tilde{z}$, given $\{0 < \tilde{z}\}$, is uniformly distributed, so

$$\begin{aligned} \text{Prob}(\tilde{M}_1(p) < \tilde{z}/Z) &= \sum_{z \in \mathcal{Z}^+} \text{Prob}(z = \tilde{z}) \text{Prob}(\tilde{M}_1(p) < z/Z) \\ &= \frac{\text{Prob}(0 < \tilde{z})}{Z} \sum_{z=1}^Z \text{Prob}(\tilde{M}_1(p) < z/Z) = \text{Prob}(0 < \tilde{z})(1 - M_1(p)) \end{aligned}$$

where the first equality holds because $\tilde{M}_1(p)$ is a nonnegative random variable.

Inserting the result into the Nash condition, we obtain

$$\forall p \in (\text{Ask}^c, V), \quad M_1(p) = 1 + \frac{\alpha(p)}{\beta(p) \text{Prob}(0 < \tilde{z})}$$

where Ask^c is given by (4).

Of course, if $Z = 1$, then Example 2 is reduced to Example 1 with $m = 1$.

6 An Extension

We relax the assumption that $\tilde{z}$ is an integer valued random variable. Instead, we assume $\tilde{z}$ can be embedded in a grid. Specifically, we replace property (1) with

$$\exists \tau_{ick} \in \mathbb{R}^{++} \text{ such that } \mathcal{Z}^+ \subset \{i \times \tau_{ick} | i \in \mathbb{N}\} \quad (1')$$

¹⁴Summation by parts implies that for any nonnegative integer random variable $\tilde{x}$, $E\tilde{x} = \sum_{i=1}^{\infty} \text{Prob}(\tilde{x} \geq i)$. Here, we use this result with the random variable $Z\tilde{M}_1(p)$.

The assumption that $\tilde{z}$ is an integer random variable corresponds to the special case $\tau_{ick} = 1$. The following adjustment needs to be made to accommodate the change: Replace (5) with

$$\begin{aligned} n_m &\equiv mZ/\tau_{ick} \\ \Delta_m &\equiv \tau_{ick}/m \end{aligned} \tag{5'}$$

The rest of this paper (definitions, theorems, lemmas, and proofs) is unchanged.

Because every finite, continuously distributed random variable $\tilde{z}$ can be approximated using a discrete random variable, we can use the relaxed assumption (1') to conclude that a Nash equilibrium is also arbitrarily close to the competitive supply function when $\tilde{z}$ is continuously distributed.¹⁵ We illustrate the convergence using the uniform example (Example 2). Assume we are interested in approximating a model in which the noise trader is equally likely to be a buyer or a seller and the conditional distribution of the noise order, given that the order is positive, is a continuous random variable with uniform distribution on the interval $(0, 1)$. Solving the tail condition yields

$$\forall q < 0 \leq 1 \quad q = 1 + \frac{\alpha(P^c(q))}{\beta(P^c(q)) \times 1/2}$$

Let $S^c(\cdot) : [\text{Ask}^c, P^c(1)]$ be

$$S^c(p) = 1 + \frac{\alpha(p)}{\beta(p) \times 1/2}$$

Then, for $q \in (0, 1]$, $P^c(q)$ is the inverse of $S^c(p)$, and $S^c(p)$ is the total number of units offered in the competitive equilibrium at prices lower than or equal to p .

Consider an approximation of the continuous standard uniform random variable: Set $\tau_{ick} = 1/10^k$ for some large integer k , and assume that $\text{Prob}(0 < \tilde{z}) = 1/2$ and the conditional distribution of $\tilde{z}$, given $\{0 < \tilde{z}\}$, is

$$U\{\tau_{ick}, 2\tau_{ick}, \dots, 10^k \times \tau_{ick}\}$$

¹⁵This is analogous to the notion that a no-arbitrage argument applied in a binomial world can be used to approximate derivative securities in the continuous economy of Black and Scholes.

Then, $Z = 1$, $n_1 = m10^k$, and $\Delta_m = 1/(m10^k)$. When $m = 1$, we can compute the mixing distribution explicitly as in Example 2, above, yielding

$$p \in [\text{Ask}^c, V], \quad 1 + \frac{\alpha(p)}{\beta(p) \times 1/2}$$

which, for $p \in [\text{Ask}^c, P^c(1)]$, is exactly $S^c(p)$.

The total number of units offered in the Nash equilibrium is

$$\frac{1}{10^k} \sum_{i=1}^{10^k+1} I_{\{\tilde{p}_i \leq p\}} \approx \text{the mixing distribution} = S^c(p)$$

where for the approximation we use the LLN.

7 Conclusion

The tail condition is a pillar of the literature of limit order markets. In this paper, we provide a strategic foundation for this condition. The equilibrium we study is in mixed strategies. A known criticism of a mixed strategies equilibrium is the lack of “no regret,” where, after a constellation of quotes is realized, suppliers would like to revise their quotes. But the frequent revision of quotes is exactly what occurs in limit order book markets, as orders are canceled and replaced in milliseconds.¹⁶ Thus, a repeated game interpretation of this paper suggests that frequent order cancellations can be part of an equilibrium and, hence, are benign rather than the result of manipulative behavior.

¹⁶Hasbrouck (2015) shows that quotes are astoundingly volatile; the number of quote changes is, on average, 20 times higher than the number of transactions, and the volatility implied by quote changes of less than 50 milliseconds is 5 times that implied by quote changes of 16 minutes.

A Appendix

Proof of Theorem 3.1, Part 1: We rewrite the Nash condition (6) in terms of distribution functions. To do so, we fix m and consider a sample of n_m Bernoulli trials, each with probability of success h . The probability that the *proportion* of successes in the random sample is strictly smaller than z/Z is the same as the probability that the *count* of successes in the random sample is strictly smaller than $n_m z/Z$. The latter probability equals $B(n_m z/Z - 1; n_m, h)$, where $B(k; n, h)$ is the distribution function of a binomial random variable with parameters n and h .¹⁷ Thus, (6) can be written as

$$\forall p \in [\text{Ask}^c, V], \quad \alpha(p) + \beta(p) \sum_{z \in \mathcal{Z}^+} \text{Prob}(z = \tilde{z}) B(n_m z/Z - 1; n_m, M_m(p)) = 0 \quad (6')$$

Our goal is to show that we can define a continuous and strictly increasing distribution function, $M_m(\cdot)$, with support $[\text{Ask}^c, V]$, that satisfies (6') in its support.

Define the function $H : [0, 1] \rightarrow [0, \text{Prob}(0 < \tilde{z})]$ via

$$H(h) = \sum_{z \in \mathcal{Z}^+} \text{Prob}(z = \tilde{z}) B(n_m z/Z - 1; n_m, h)$$

Then, $H(\cdot)$ is continuous and strictly decreasing with¹⁸

$$H(0) = \text{Prob}(0 < \tilde{z})$$

$$H(1) = 0$$

Therefore, $H^{-1} : [0, \text{Prob}(0 < \tilde{z})] \rightarrow [0, 1]$ exists, it is continuous and strictly decreasing, and it satisfies the boundary conditions

¹⁷ $B(k; n, h)$ is the probability that out of n trials, there are at most k successes.

¹⁸Informally, when we increase h , the probability of success in each trial, the probability of having at most x strictly decreases. Formally, we write

$$B(k; n, h) = (n - k) \binom{n}{k} \int_0^{1-h} t^{n-k-1} (1-t)^k dt$$

and simply differentiate with regard to h .

$$H^{-1}(\text{Prob}(0 < \tilde{z})) = 0$$

$$H^{-1}(0) = 1$$

Define

$$M_m(p) := H^{-1}\left(\frac{-\alpha(p)}{\beta(p)}\right)$$

By construction, $M_m(p)$ satisfies (6').

We need to verify that $M_m : [\text{Ask}^c, V] \rightarrow [0, 1]$ is a continuous and strictly increasing function with $M_m(\text{Ask}^c) = 0$ and $M_m(V) = 1$.

From the definition of Ask^c , we find that $-\alpha(\text{Ask}^c)/\beta(\text{Ask}^c) = \text{Prob}(0 < \tilde{z})$. Therefore, $M_m(\text{Ask}^c) = 0$. Because $\alpha(V) = 0$, we also have $M_m(V) = 1$.

The continuity of $H^{-1}(\cdot)$, $\alpha(\cdot)$, and $\beta(\cdot)$, and the fact that in $[\text{Ask}^c, V]$, $\beta(\cdot) > 0$, implies the continuity of $M_m(\cdot)$.

To see that $p \rightarrow M_m(p)$ is strictly increasing, note that in $[\text{Ask}^c, V]$ the function $p \rightarrow -\alpha(p)/\beta(p)$ is strictly decreasing. Because $H^{-1}(\cdot)$ is also strictly decreasing, it follows that $M_m(\cdot)$ is strictly increasing.

□

Proof of Theorem 3.1, Part 2a. Let $P(\cdot)$ be an arbitrary price schedule that the active trader may use. Let $\tilde{q}$ be the number of units the trader sells when posting $P(\cdot)$, whereas all other traders follow their conjectured equilibrium strategies. In particular, n_m active liquidity suppliers offer Δ_m at i.i.d. random prices with distribution function $M_m(\cdot)$.¹⁹

¹⁹Because $M_m(\cdot)$ is continuously increasing, $\text{prob}(\tilde{p}_j \leq p) = \text{Prob}(\tilde{p}_j < p)$. For definiteness, in this proof we use the strict inequality version.

For every $q > 0$,

$$\begin{aligned}
\{q \leq \tilde{q}\} &= \{q \leq \tilde{q}, \text{demander is informed}\} \cup \{q \leq \tilde{q}, \text{demander is noise}\} \\
&= \left\{ \{\text{demander is informed}\} \cap \{P(q) < \tilde{v}\} \right\} \\
&\quad \cup \left\{ \{\text{demander is noise}\} \cap \left\{ q + \sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right\} \right\}
\end{aligned}$$

The four events are disjoint. Thus, the active trader's expected payoff is

$$\begin{aligned}
&E \int_0^\infty I_{\{q \leq \tilde{q}\}} (P(q) - \tilde{v}) dq \\
&= \int_0^\infty \mu E I_{\{P(q) < \tilde{v}\}} (P(q) - \tilde{v}) dq + (1 - \mu) \text{Prob} \left(q + \sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right) (P(q) - \tilde{v}) dq \\
&= \int_0^\infty \alpha(P(q)) + \beta(P(q)) \text{Prob} \left(q + \sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right) dq
\end{aligned}$$

Ignoring the constraint that $q \rightarrow P(q)$ must be nondecreasing, we can solve this optimization problem pointwise (for each q , let $P(q)$ maximize the integrand). If an argument of maximum of the unconstrained problem is a nondecreasing price schedule, then it is also a solution for the constraint problem.

Therefore, for every q , we want to find a value of p that maximizes the integrand

$$\alpha(p) + \beta(p) \text{Prob} \left(q + \sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right) \quad (8)$$

Fix $q \in (0, \Delta_m]$. We have

$$\begin{aligned}
\text{Prob} \left(q + \sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < p\}} \leq \tilde{z} \right) &= \text{Prob} \left(\Delta_m + \sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < p\}} \leq \tilde{z} \right) \\
&= \text{Prob} \left(\sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < p\}} < \tilde{z} \right) \\
&= \text{Prob} \left(\frac{1}{n_m} \sum_{j=1}^{n_m} I_{\{\tilde{p}_j < p\}} < \tilde{z}/Z \right) = \text{Prob} \left(\tilde{M}_m(p) < \tilde{z}/Z \right)
\end{aligned}$$

Here, the first and second equalities arise because $\mathcal{Z}^+ \subset \{i\Delta_m | i \in \mathbb{N}\}$. The third and fourth equalities follow from the definitions of n_m and $\tilde{M}_m(p)$, respectively.

Thus, for $q \in (0, 1/m]$, the integrand (8) equals

$$\alpha(p) + \beta(p) \text{Prob} \left(\tilde{M}_m(p) < \tilde{z}/Z \right) \leq 0$$

with equality only if $p \geq \text{Ask}^c$.²⁰

We conclude that for each $q \in (0, 1/m]$, any $p \geq \text{Ask}^c$ maximizes the integrand and the maximum is zero. In particular, $p \in [\text{Ask}^c, V]$ is optimal.

Because the integrand (8) is decreasing in q , as well as for $q > \Delta_m$, it cannot be strictly positive.

In summary, we have shown that the maximum expected profit is zero and that it is optimal to offer any q , provided $q \leq \Delta_m$, at any price in the support of $M_m(p)$. In particular, it is optimal to offer the entire “block” of Δ_m at a random price with distribution $M_m(p)$.

□

Proof of Theorem 3.1, Part 2b. Let $P(\cdot)$ be an arbitrary price schedule that the nonactive trader may use. Let $\tilde{q}$ be the number of units the trader sells when posting $P(\cdot)$, whereas all other traders follow their conjectured equilibrium strategies. In particular, each of the $n_m + 1$ active liquidity suppliers offers Δ_m at a random price that has the distribution function $M_m(\cdot)$.

²⁰The definition of Ask^c and the monotonicity of $\alpha(\cdot)$ and $\beta(\cdot)$ imply that for $p < \text{Ask}^c$ the integrand is negative.

For every $q > 0$,

$$\begin{aligned} \{q \leq \tilde{q}\} &= \{q \leq \tilde{q}, \text{demander is informed}\} \cup \{q \leq \tilde{q}, \text{demander is noise}\} \\ &= \left\{ \{\text{demander is informed}\} \cap \{P(q) < \tilde{v}\} \right\} \\ &\quad \cup \left\{ \{\text{demander is noise}\} \cap \left\{ q + \sum_{j=1}^{n_m+1} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right\} \right\} \end{aligned}$$

The four events are disjoint. Thus, the nonactive trader's expected payoff is

$$\begin{aligned} &E \int_0^\infty I_{\{q \leq \tilde{q}\}} (P(q) - \tilde{v}) dq \\ &= \int_0^\infty \mu E I_{\{P(q) < \tilde{v}\}} (P(q) - \tilde{v}) dq + (1 - \mu) \text{Prob} \left(q + \sum_{j=1}^{n_m+1} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right) (P(q) - \tilde{v}) dq \\ &= \int_0^\infty \alpha(P(q)) + \beta(P(q)) \text{Prob} \left(q + \sum_{j=1}^{n_m+1} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right) dq \end{aligned}$$

Ignoring the constraint that $P(q)$ must be nondecreasing, we can solve this optimization problem pointwise (for each q , let $P(q)$ maximize the integrand). If an argument of maximum of the unconstrained problem is a nondecreasing price schedule, then it is also a solution for the constraint problem.

Therefore, for every q , we want to find a value of p that maximizes the integrand

$$\alpha(p) + \beta(p) \text{Prob} \left(q + \sum_{j=1}^{n_m+1} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right)$$

Fix $q > 0$. If $p \leq E\tilde{v}$, then $\beta(p)$ is negative and the integrand is negative. For $p > E\tilde{v}$, we have

$$\begin{aligned} &\alpha(p) + \beta(p) \text{Prob} \left(q + \sum_{j=1}^{n_m+1} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right) \\ &\leq \alpha(p) + \beta(p) \text{Prob} \left(\sum_{j=1}^{n_m} \Delta_m I_{\{\tilde{p}_j < P(q)\}} \leq \tilde{z} \right) \leq \alpha(p) + \beta(p) \text{Prob} \left(\tilde{M}_m(p) < \tilde{z}/Z \right) \leq 0 \end{aligned}$$

The last inequality is the Nash condition (6). Thus, a nonactive trader has no profitable deviation. $\square$

Proof of Lemma 4.1. Fix p , and let $M_{m_k}(p)$ have a convergent subsequence.

For every $m \geq 1$, we have

$$\begin{aligned} E\tilde{M}_m(p) &= M_m(p) \\ \text{Var}(\tilde{M}_m(p)) &= \frac{M_m(p)(1 - M_m(p))}{n_m} \end{aligned}$$

For every $\epsilon > 0$, Chebyshev inequality implies

$$\begin{aligned} \text{Prob}(|\tilde{M}_m(p) - M_m(p)| > \epsilon) &\leq \frac{\text{Var}(\tilde{M}_m(p))}{\epsilon^2} \\ &= \frac{M_m(p)(1 - M_m(p))}{n_m \epsilon^2} \leq \frac{1}{4n_m \epsilon^2} \xrightarrow{m \rightarrow \infty} 0 \end{aligned}$$

Thus, $\tilde{M}_m(p) - M_m(p)$ converges to zero in probability; the subsequence $\tilde{M}_{m_k}(p) - M_{m_k}(p)$ also converges to zero in probability. The latter statement implies that the subsequence $\tilde{M}_{m_k}(p)$ converges in probability to the same limit of the converging subsequence $M_{m_k}(p)$.²¹

Convergence in probability implies convergence in distribution. Therefore, this completes the proof. $\square$

Proof of Lemma 4.2. Fix $q \in (0, Z)$ with $q \notin \mathcal{Z}^+$. For $p \notin [\text{Ask}^c, V]$, the statement is trivially true.²² Thus, for the remainder of this proof, p is in the common support; in other words, $p \in [\text{Ask}^c, V]$. We divide the proof into the two relevant cases: $p < P^c(q)$ and $p > P^c(q)$.

Case $p < P^c(q)$

Because $\alpha(\cdot)$ is increasing and $\beta(\cdot)$ is strictly increasing, by replacing $P^c(q)$ with p in the tail condition (3) we obtain

$$0 > \alpha(p) + \beta(p) \text{Prob}(q \leq \tilde{z}) \tag{9}$$

²¹The limit of the sum equals the sum of limits. Here, the two sequences are $\tilde{M}_{m_k}(p) - M_{m_k}(p)$ and $M_{m_k}(p)$, and their sum is $\tilde{M}_{m_k}(p)$.

²²For $p < \text{Ask}^c \leq P^c(q)$, $\tilde{M}_m(p) \equiv 0$. Because $q > 0$, the statement is true. For $p > V > P^c(q)$, $\tilde{M}_m(p) \equiv 1$. Because $q < Z$, the statement is true.

Our goal is to show that $\lim \text{Prob}(\tilde{M}_m(p) \leq q/Z) = 1$. If we can show that the limit inferior of the sequence $\text{Prob}(\tilde{M}_m(p) \leq q/Z)$ is one, then we are done. Take a subsequence that converges to the limit inferior. Any further subsequence converges to the same limit. We extract a further subsequence such that the mixing distribution converges. In summary, $M_{m_k}(p)$ converges to some limit and $\text{Prob}(\tilde{M}_{m_k}(p) \leq q/Z)$ converges to the limit inferior of the sequence $\text{Prob}(\tilde{M}_m(p) \leq q/Z)$. If the limit of $M_{m_k}(p)$ is strictly smaller than q/Z , then Lemma 4.1 implies $\text{Prob}(\tilde{M}_{m_k}(p) \leq q/Z) \rightarrow 1$, and we are done. Assume, therefore, by means of contradiction, that

$$\lim M_{m_k}(p) \geq q/Z$$

For every $z \in \mathcal{Z}^+$, with $q > z$, we have $\lim M_{m_k}(p) > z/Z$. Therefore, Lemma 4.1 implies that $\text{Prob}(\tilde{M}_{m_k}(p) \leq z/Z) \rightarrow 0$. From the Nash condition (6), we have

$$\begin{aligned}
0 &= \alpha(p) + \beta(p) \text{Prob}(\tilde{M}_{m_k}(p) < \tilde{z}/Z) \\
&= \alpha(p) + \beta(p) \sum_{z \in \mathcal{Z}^+} \text{Prob}(\tilde{z} = z) \text{Prob}(\tilde{M}_{m_k}(p) < z/Z) \\
&\leq \alpha(p) + \beta(p) \left(\sum_{\substack{q > z \\ z \in \mathcal{Z}^+}} \text{Prob}(\tilde{z} = z) \text{Prob}(\tilde{M}_{m_k}(p) \leq z/Z) + \sum_{\substack{q \leq z \\ z \in \mathcal{Z}^+}} \text{Prob}(\tilde{z} = z) \right) \\
&\qquad\qquad\qquad \downarrow \qquad\qquad\qquad \parallel \\
&\qquad\qquad\qquad 0 \qquad\qquad\qquad \text{Prob}(q \leq \tilde{z}) \\
&\longrightarrow \alpha(p) + \beta(p) \text{Prob}(q \leq \tilde{z})
\end{aligned}$$

which contradicts (9), and concludes the proof for the case $p < P^c(q)$.

Case $p > P^c(q)$ ²³

²³This is the part of the proof where we need to use the assumption that $q \notin \mathcal{Z}^+$.

Because $\alpha(\cdot)$ is increasing and $\beta(\cdot)$ is strictly increasing, by replacing $P^c(q)$ with p in the tail condition (3) we obtain

$$0 < \alpha(p) + \beta(p) \text{Prob}(q \leq \tilde{z}) \quad (10)$$

Our goal is to show that $\lim \text{Prob}(\tilde{M}_m(p) \leq q/Z) = 0$. If we can show that the limit superior of the sequence $\text{Prob}(\tilde{M}_m(p) \leq q/Z)$ is zero, then we are done. Take a subsequence that converges to its limit superior. Any further subsequence converges to the same limit. We extract a further subsequence such that the mixing distribution converges. In summary, $M_{m_k}(p)$ converges to some limit and $\text{Prob}(\tilde{M}_{m_k}(p) \leq q/Z)$ converges to the limit superior of the sequence $\text{Prob}(\tilde{M}_m(p) \leq q/Z)$. If the limit of $M_{m_k}(p)$ is strictly greater than q/Z , then Lemma 4.1 implies $\text{Prob}(\tilde{M}_{m_k}(p) \leq q/Z) \rightarrow 0$, and we are done. Assume, therefore, by means of contradiction, that

$$\lim M_{m_k}(p) \leq q/Z$$

For every $z \in \mathcal{Z}^+$ with $q < z$, there is a q' such that $q < q' < z$. Therefore, $\text{Prob}(\tilde{M}_{m_k}(p) \leq q'/Z) \leq \text{Prob}(\tilde{M}_{m_k}(p) < z/Z)$. By contrast, $q < q'$, so $\lim M_{m_k}(p) < q'/Z$. Therefore, Lemma 4.1 implies that

$$1 \leftarrow \text{Prob}(\tilde{M}_{m_k}(p) \leq q'/Z) \leq \text{Prob}(\tilde{M}_{m_k}(p) < z/Z) \leq 1$$

Next, from the Nash condition (6), we have

$$\begin{aligned}
0 &= \alpha(p) + \beta(p) \sum_{z \in \mathcal{Z}^+} \text{Prob}(\tilde{z} = z) \text{Prob}(\tilde{M}_{m_k}(p) < z/Z) \\
&\geq \alpha(p) + \beta(p) \sum_{q < z, z \in \mathcal{Z}^+} \text{Prob}(\tilde{z} = z) \text{Prob}(\tilde{M}_{m_k}(p) < z/Z) \\
&\hspace{25em} \downarrow \\
&\hspace{25em} 1 \\
&\longrightarrow \alpha(p) + \beta(p) \text{Prob}(q < \tilde{z}) \\
&= \alpha(p) + \beta(p) \text{Prob}(q \leq \tilde{z})
\end{aligned}$$

where the last equality is true because $q \notin \mathcal{Z}^+$. This contradicts (10), and concludes the proof for the case $p > P^c(q)$.

□

Proof of Theorem 4.3: Fix $q \in (0, Z)$ with $q \notin \mathcal{Z}^+$. We divide the proof into the two relevant cases: $p > P^c(q)$ and $p < P^c(q)$.

Case $p > P^c(q)$

We need to show that $\text{Prob}(\tilde{P}^m(q) \leq p) \rightarrow 1$. We have

$$\begin{aligned}
&\text{Prob}(\tilde{P}^m(q) \leq p) \geq \text{Prob}(\tilde{M}_m(p) \geq q/Z) \geq \text{Prob}(\tilde{M}_m(p) > q/Z) \\
&= 1 - \text{Prob}(\tilde{M}_m(p) \leq q/Z) \longrightarrow 1 \\
&\hspace{10em} \downarrow \\
&\hspace{10em} 0
\end{aligned}$$

The first inequality is the left inequality of (7). The second inequality is trivial. For the limit, we use Lemma 4.2.

Case $p < P^c(q)$

We need to show that $\text{Prob}(\tilde{P}^m(q) \leq p) \rightarrow 0$. Because $q \notin \mathcal{Z}^+$, in a small neighborhood of q , $P^c(\cdot)$ is flat. Thus, there is a small $a > 0$ such that (i) $q - a \in (0, Z)$ with $q - a \notin \mathcal{Z}^+$, and (ii) $P^c(q) = P^c(q - a)$.

For sufficiently large m , we have $1/n_m < a/Z$. Therefore,

$$\begin{aligned} \text{Prob}(\tilde{P}^m(q) \leq p) &\leq \text{Prob}\left(\tilde{M}_m(p) \geq q/Z - \frac{1}{n_m}\right) \leq \text{Prob}\left(\tilde{M}_m(p) > \frac{q-a}{Z}\right) \\ &= 1 - \text{Prob}\left(\tilde{M}_m(p) \leq \frac{q-a}{Z}\right) \rightarrow 0 \\ &\quad \downarrow \\ &\quad 1 \end{aligned}$$

The first inequality is the right inequality of (7). The second equality holds because we are considering only large values of m such that $1/n_m < a/Z$. For the limit, we use Lemma 4.2 at price p and quantity $q - a$, knowing that $p < P(q) = P^c(q - a)$.

□

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